Web Design Exam Review Guide

**145010 — Web Design | Web Exam Prep

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1. Internet Law, Ethics & Professional Practice

Creative Commons — A set of public copyright licenses that allow free distribution of an otherwise copyrighted work. Creators choose how others may use their content (share, remix, commercial use, etc.).

Intellectual Property — Any work that is the result of creativity to which the creator has legal rights. You can apply for a copyright, patent, or trademark to protect it.

Acceptable Use Policy (AUP) — A set of rules defining how a network, system, or website may be used. Schools, businesses, and ISPs all use AUPs.

Anti-Spam Laws (CAN-SPAM Act) — Require commercial emails to clearly identify themselves as advertisements, provide an opt-out option, and include a valid sender address.

Privacy Laws — Laws that prohibit the disclosure or misuse of information about private individuals. Data protection regulations control how personal data is collected, stored, and shared.

Netiquette — The correct or acceptable way of communicating on the Internet — things like using appropriate language, not typing in ALL CAPS, and respecting others' time and bandwidth.

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2. HTML Fundamentals

HTML (Hypertext Markup Language) — The standard coding language used to structure and format content for the World Wide Web. Browsers read HTML to display web pages.

Elements — Everything from a start tag to its end tag, including the content in between.

```
<p>This entire line is an element.</p>
```

Tags — The HTML markup itself, from `<` to `>`. Tags wrap around content to create elements.

Attributes — Provide additional information about an element using name/value pairs.

```
<img src="photo.jpg" alt="A sunset">
```

Semantic vs. Non-Semantic Elements

Semantic elements clearly describe their meaning to both the browser and the developer. They improve accessibility and SEO.

Semantic	Purpose
`<header>`	Introductory content or navigation container
`<nav>`	Set of navigation links
`<main>`	Primary content of the page
`<section>`	Thematic grouping of content
`<article>`	Independent, self-contained content (blog post, news story)
`<aside>`	Content tangentially related to surrounding content (sidebar)
`<footer>`	Author info, copyright, contact links, sitemap

Non-semantic elements tell nothing about their content. Examples: `<div>` (division/section) and `<span>` (inline container). Use these when no semantic tag fits.

Block vs. Inline Elements

Block-level elements start on a new line and take up the full width available. Examples: `<div>`, `<h1>`–`<h6>`, `<p>`, `<section>`.

Inline elements do not start on a new line and only take up as much width as needed. Examples: `<span>`, `<a>`, `<img>`, `<strong>`.

Headings

`<h1>` is the largest heading and should be used once per page for the main title. Headings range from `<h1>` (largest) down to `<h6>` (smallest).

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3. Links, Media & Special Characters

Absolute vs. Relative Links

Absolute link — A complete URL including the protocol. Used for external sites.

```
<a href="https://www.example.com/page">Visit Example</a>
```

Relative link — A path referencing a file on the same server. No protocol needed.

```
<a href="about.html">About Us</a>
```

Link Pseudo-Classes (CSS)

Selector	What It Styles

Selector	What It Styles
`a:link`	Unvisited link
`a:visited`	Previously visited link
`a:hover`	Link when mouse is over it
`a:active`	Link at the moment it is clicked

Character Entities (Special Characters)

Used to display reserved characters in HTML that would otherwise be interpreted as code.

Entity	Displays
`&`	&
` `	Non-breaking space
`<`	<
`>`	>
`©`	(c)

Meta Elements

Reside in the `<head>` section. They provide metadata about the page — keywords, descriptions, author, and viewport settings — that help with SEO and responsiveness.

Keywords — Words or phrases describing page content, used as metadata for search engines and image descriptions.

Viewport — The user's visible area of a web page. It varies by device. Setting the viewport meta tag ensures proper scaling on mobile.

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

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4. CSS (Cascading Style Sheets)

CSS controls a website's text color, backgrounds, borders, spacing, positioning, and more.

Three Ways to Apply CSS

Inline style — Applied directly to a single element using the `style` attribute. Highest specificity.

```
<p style="color: blue;">Blue text</p>
```

Internal style — Written inside `<style>` tags in the `<head>`. Styles one page.

```
<style>  p { color: blue; }  </style>
```

External style sheet — A separate `.css` file linked in the `<head>`. Best practice — one file can style an entire website.

```
<link rel="stylesheet" href="styles.css">
```

Cascade Order (When Rules Conflict)

Priority from highest to lowest: **Inline > Internal > External**. The most specific rule wins.

Key CSS Concepts

ID selector (`#name`) — Used only once per page. Targets a unique element.

Class selector (`.name`) — Can be used multiple times on a page. Targets groups of elements.

Font attributes — type (font-family), size (font-size), color.

Background attributes — color and image.

Border attributes — width, style, and color.

```
border: 2px solid #333;
```

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5. Web Design Principles

Validation (W3C) — The process of checking that your HTML/CSS conforms to web standards defined by the World Wide Web Consortium. Ensures cross-browser consistency and accessibility.

Usability — How easy a user interface is to learn and use. Good usability means users can accomplish tasks efficiently.

Readability — How easily a person can understand your content. Affected by font choice, size, line spacing, contrast, and content complexity.

Accessibility — The practice of ensuring websites can be used by everyone, including people with physical disabilities, situational limitations, and limited bandwidth. Related standards include the Americans with Disabilities Act (ADA).

Cross-Browser Compatibility — Testing and debugging your site across different browsers and devices so it works consistently everywhere.

Wireframe — A visual prototype of a webpage focusing on content placement, layout, and behavior — not colors or final design.

Site Map — A structured list of all pages on a website, showing how they connect.

Image Resolution / Optimization — Optimizing images so they load quickly while maintaining acceptable quality. Higher resolution means more detail but larger file size.

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6. Web Development Roles

Front-End Developer — Builds what users see and interact with, using HTML, CSS, and JavaScript.

Back-End Developer — Works with servers, databases, and dynamic content using languages like PHP.

UX (User Experience) Designer — Focused on how the product *feels*. Creates wireframes, prototypes, and conducts user testing.

UI (User Interface) Designer — Focused on how the product *looks*. Handles buttons, icons, spacing, typography, color schemes, and responsive design.

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7. Web Forms (Outcome 6.4)

Forms collect user input using a variety of elements: text fields, radio buttons, check boxes, drop-down menus, and submit/reset buttons.

Form action — The `action` attribute specifies where form data is sent when submitted.

```
<form action="process.php" method="post">
```

Key form elements to know: `<input>`, `<select>`, `<textarea>`, `<button>`, `<fieldset>`, `<label>`.

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8. Quick-Reference: Document Structure

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Page Title (shows in browser tab)</title>
    <link rel="stylesheet" href="styles.css">
  </head>
  <body>
    <header>Site header / navigation</header>
    <main>
      <section>
        <article>Main content here</article>
      </section>
```

```
        <aside>Sidebar content</aside>
      </main>
      <footer>Copyright, contact info</footer>
    </body>
  </html>
```

Best Coding Practice — Informal rules adopted by the development community to keep code clean, maintainable, and consistent. Using external CSS, semantic HTML, proper indentation, and W3C validation are all best practices.